CHAPTER 1: INFORMED SEARCH & ADVERSARIAL ALGORITHMS (Page 1)

[Chapter 1: Search & Logic • Section 1.1 • Page 1]
Search algorithms navigate state spaces from an initial state to a goal state.
Uninformed search strategies (BFS, DFS) explore without domain knowledge. BFS expands the shallowest node using a FIFO queue, guaranteeing completeness and optimality for unit-cost step problems, with time and space complexity O(b^d). DFS expands the deepest unexpanded node using a LIFO stack, with space complexity O(b*m), but is not optimal and can trap in infinite loops without cycle detection.

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Informed (Heuristic) search leverages a heuristic function h(n) estimating the cheapest path from node n to the goal.
The A* search algorithm evaluates nodes using f(n) = g(n) + h(n), where g(n) is the exact cost incurred from the start node to n, and h(n) is the estimated cost from n to the goal.
A heuristic h(n) is admissible if it never overestimates the true cost to reach the goal: 0 <= h(n) <= h*(n).
A heuristic is consistent (monotonic) if for every node n and successor n' generated by action a: h(n) <= c(n, a, n') + h(n') and h(Goal) = 0.
Theorem: If h(n) is admissible on tree search, or consistent on graph search, A* is guaranteed to be complete and optimal.

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In two-player zero-sum games, the Minimax algorithm computes optimal decisions assuming an adversarial opponent.
Alpha-Beta pruning eliminates subtrees that cannot influence the final decision without losing optimality:
- Alpha (alpha): The highest-value choice found so far along any choice point for MAX. Initialized to -infinity.
- Beta (beta): The lowest-value choice found so far along any choice point for MIN. Initialized to +infinity.
A branch is pruned whenever alpha >= beta. In the ideal move-ordering case, alpha-beta pruning reduces the effective branching factor from b to sqrt(b), doubling the reachable search depth.

[Chapter 2: Deep Learning • Section 2.1 • Page 2]
Artificial Neural Networks learn non-linear representations by stacking parameterized affine transformations interleaved with element-wise non-linear activations: $z^{(l)} = W^{(l)} * a^{(l-1)} + b^{(l)}$, with $a^{(l)} = \sigma(z^{(l)})$.
Training minimizes an objective loss $L(\theta)$ via reverse-mode automatic differentiation (Backpropagation). Backpropagation recursively applies the multivariate chain rule from the final loss backward toward input weights. For weight matrix $W^{(l)}$, the gradient is computed as: $\frac{dL}{dW^{(l)}} = \delta^{(l)} * (a^{(l-1)})^T$, where $\delta^{(l)} = \frac{dL}{dz^{(l)}}$ represents the local error vector.

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The choice of activation function governs gradient flow across deep networks.
The Sigmoid function, $\sigma(z) = 1 / (1 + \exp(-z))$, squashes inputs to (0, 1). Its derivative, $\sigma'(z) = \sigma(z) * (1 - \sigma(z))$, has a maximum value of only 0.25 at $z = 0$. For large absolute values of z, the derivative approaches zero, causing the vanishing gradient problem where early layers fail to update.
The Rectified Linear Unit, $ReLU(z) = \max(0, z)$, has a constant derivative of 1 for all positive inputs, allowing gradient signals to propagate across dozens of layers without exponential decay. However, neurons with $z < 0$ have zero gradient and can permanently deactivate (the "Dying ReLU" phenomenon), which is mitigated by Leaky ReLU, $f(z) = \max(0.01z, z)$, or GELU (Gaussian Error Linear Unit).

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Modern deep networks are optimized using adaptive moment estimation.
Adam maintains exponentially decaying moving averages of past gradients (first moment m_t, estimating the mean) and past squared gradients (v_t, estimating uncentered variance):
$m_t = \beta_1 * m_{t-1} + (1 - \beta_1) * g_t$
$v_t = \beta_2 * v_{t-1} + (1 - \beta_2) * (g_t)^2$
AdamW (Loshchilov & Hutter) corrects standard Adam by decoupling L2 weight decay regularization from the gradient update step, preventing adaptive scale distortions on penalization weights and improving test generalization.

[Chapter 3: Transformers • Section 3.1 • Page 3]
The Transformer architecture (Vaswani et al., 2017) dispenses with recurrent recurrences entirely, computing token dependencies in parallel via Self-Attention. Given an input sequence mapped to embedding matrix X in R^(n x d_model), linear projections yield Queries (Q = X * W_Q), Keys (K = X * W_K), and Values (V = X * W_V), where W_Q, W_K in R^(d_model x d_k) and W_V in R^(d_model x d_v).
Scaled Dot-Product Attention is formulated as:
Attention(Q, K, V) = softmax((Q * K^T) / sqrt(d_k)) * V
The scaling factor 1 / sqrt(d_k) prevents the dot products from growing excessively large for high dimensions d_k, which would push the softmax function into regions with near-zero gradients.

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Multi-Head Attention (MHA) projects queries, keys, and values h times into different representation subspaces:
MultiHead(Q, K, V) = Concat(head_1, ..., head_h) * W_O
where head_i = Attention(Q * W_Q^(i), K * W_K^(i), V * W_V^(i)). This allows the model to simultaneously attend to information from different representation subspaces at different positions.

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In decoder-only autoregressive LLMs (like GPT and LLaMA), causal masking is applied to ensure prediction of token at position t depends only on tokens at indices <= t. During inference, recomputing all key and value tensors for prior tokens at every generation step scales quadratically O(t^2). The KV-Cache retains key and value projections for past tokens in GPU memory, reducing step generation complexity to O(t). Rotary Position Embeddings (RoPE) encode relative token distances by rotating the query and key vectors in complex 2D planes, preserving relative distance decay across long context windows.

[Chapter 4: RAG Systems • Section 4.1 • Page 4]
Retrieval-Augmented Generation (RAG) decouples parametric model memory from non-parametric factual knowledge.
An LLM trained on a static corpus cannot recall private enterprise data and suffers from factual hallucinations when probed beyond its training cutoff.
RAG implements a strict two-stage pipeline:
1. Ingestion Phase: Documents are cleaned, segmented into discrete chunks (e.g., 300 to 500 words with 10% overlap), and mapped into dense vector embeddings using an embedding model (e.g., text-embedding-3-small, BAAI/bge-large).
2. Retrieval Phase: A user query is embedded into the same vector space. A similarity metric (Cosine Similarity = (u . v) / (||u|| * ||v||)) retrieves the Top-K closest passages from an Approximate Nearest Neighbor index (such as Hierarchical Navigable Small World, HNSW).
3. Augmented Generation Phase: The retrieved passages are injected into the LLM system prompt as verified context boundaries. The model is instructed to answer strictly using the provided excerpts and cite source sections.

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Retrieval Guardrails and Grounding:
To eliminate hallucinations, the retrieval module must verify context relevance before generation. If the highest similarity score between the query and the knowledge base falls below a strict threshold (e.g., Cosine Similarity < 0.10 or TF-IDF overlap < 0.08), the system flags the query as Out-of-Scope and refuses generation, rather than allowing the LLM to invent plausible-sounding false statements.

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Trade-offs: RAG vs Fine-Tuning (LoRA):
- Parameter-Efficient Fine-Tuning (LoRA / QLoRA) alters model weights by injecting low-rank decomposition matrices W = W_0 + (B * A) * (alpha / r). Fine-tuning excels at teaching tone, syntax, dialect, or strict formatting styles, but is brittle for factual recall and cannot dynamic update when files change daily.
- RAG updates instantly without gradient computation or GPU retraining simply by adding or removing documents from the vector store. RAG provides deterministic, verifiable citations that can be inspected and audited by human examiners.